SRM

DEPARTMENT OF CHEMISTRY **SET 1**
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

INTERNAL ASSESSMENT – II

Program: B. Tech.	Date: 12-11-2022
Course Code & Title: 21CYB101J & Chemistry	Duration: 12.30-01.30 PM
Year & Semester: I Year & I Semester	Max. Marks: 30 Marks

Part- A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. The change of enthalpy in an endothermic reaction is
   (a) positive (b) negative (c) constant (d) can be any of the above

2. Which of the following is an **incorrect** statement?
   (a) Corrosion is an irreversible process
   (b) Corrosion is a non-spontaneous process
   (c) Corrosion is a degradation process
   (d) Corrosion is a spontaneous process

3. Rusting of iron in acidic aqueous solution of electrolyte occurs with the evolution of _____gas.
   (a)Nitrogen
   (b)Chlorine
   (c)Oxygen
   (d) Hydrogen

4. A system absorbs 50 J of heat and does 11 J of work in a process. The system follows a different thermodynamic path between the same initial and final states and does 15 J work, the heat transferred in the process is
   (a) 39 J (b) 65 J (c) 54 J (d) 46 J

5. Which of the following is **false** regarding galvanic cells?
   (a) It converts chemical energy into electrical energy
   (b) The electrolytes taken in the two beakers are different
   (c) The reactions taking place are non-spontaneous
   (d) To set up this cell, a salt bridge is used

6. ΔG for the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ is
   (a) –FE (b) FE (c) 2FE (d) -2FE

7. For which of the following conditions a reaction will be spontaneous at all the temperature?
   (a) $\Delta H < 0$ and $\Delta S < 0$
   (b) $\Delta H < 0$ and $\Delta S = 0$
   (c) $\Delta H < 0$ and $\Delta S > 0$

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(d) both (b) and (c)

8. Which of the following is the application of Pourbaix diagrams?
(a) Predicting the spontaneous direction of reactions
(b) To reduce corrosion attack
(c) To estimate the composition of corrosion product
(d) All of the above

9. Which of the following statements is **not** correct?
(a) Fischer projection represents the molecule in an eclipsed conformation
(b) Newman projection can be represented in eclipsed, staggered and skew conformations
(c) Fischer projection of the molecule is its most stable conformation
(d) In Sawhorse projections, the lines are inclined at an angle of 120° to each other

10. The compounds $C_2H_5OC_2H_5$ and $CH_3OCH_2CH_2CH_3$ are
(a) chain isomers
(b) geometrical isomers
(c) metamers
(d) conformational isomers

Part- B (2 x 10 = 20 Marks)

11 **a. i.** What is Nernst Equation? Derive its expression and describe one application of Nernst equation in detail. (10 Marks)

OR

b. i. The cell in which the following reaction occurs:

$$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + I_{2\,(aq)}$$

has $E^\circ_{cell} = 0.236V$ at 298 K. Calculate the standard Gibbs free energy and the equilibrium constant of the reaction. (Given: F = 96,485 C/mol) (6 Marks)

ii. Differentiate between Dry Corrosion and Wet corrosion (4 Marks)

12 **a. i.** What is Gibb's free energy? Derive its expression. (6 Marks)

ii. Define entropy and give its significance. (4 Marks)

OR

b. i. Write a note on structural isomerism in organic compounds. (6 Marks)

ii. Draw Newman projections for the eclipsed and staggered conformations of ethane. Which of these conformations is more stable and why? (4 Marks)

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DEPARTMENT OF CHEMISTRY **SET 2**
College of Engineering and Technology
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INTERNAL ASSESSMENT – II

Program: B. Tech.	Date: 12-11-2022
Course Code & Title: 21CYB101J & Chemistry	Duration: 12.30-01.30 PM
Year & Sem: I Year & I Sem	Max. Marks: 30 Marks

Part- A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. Nernst equation for an electrode is based on the variation of electrode potential of an electrode with
(a) temperature (b) concentration of electrolyte (c) both a and b (c) density of electrons

2. Which of the following condition is satisfied when the cell reaction in the electrochemical cell is spontaneous?
(a) $\Delta G^\circ > 0$ (b) $E^\circ_{cell} < 0$ (c) $E^\circ_{cell} = 0$ (d) $\Delta G^\circ < 0$

3. During an adiabatic expansion of 3 moles of a gas, the change in internal energy was found to be equal to -100 J. The work done during the process is equal to:
(a) 122 J (b) 100 J (c) -300 J (d) 300 J

4. Which of the following is correct for the net entropy change in an irreversible process?
(a) It is positive (b) It is negative (c) It is zero (d) All of the above

5. Which of the following option describes the limitations of Pourbaix diagrams?
(a) Rate of reaction is can't be predicted
(b) It neglects the impurities of working conditions
(c) Stability of metallic species is predicted
(d) Rate of reaction is not predicted and it neglects the impurities of working conditions

6. Anhydrous inorganic liquid metal surface in absence of moisture undergoes _____.
(a) Wet corrosion
(b) Dry corrosion
(c) Galvanic corrosion
(d) Pitting corrosion

7. Which of the following materials will undergo corrosion?
(a) Metals only
(b) Metals and Non-metals
(c) Metals, Non-metals, Ceramics and Plastics
(d) Metals, Non-metals, Ceramics, Plastics and Rubbers

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8. In an isothermal process, the internal energy of gas molecules
(a) Increases (b) decreases (c) remains constant (d) none of these

9. The isomerism which exists between CH_3CHCl_2 and CH_2ClCH_2Cl is
(a) Chain (b) Functional (c) Position (d) geometrical isomers

10. Which of the following is true about Fischer Projection?
(a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.
(b) The vertical lines are oriented towards you and the horizontal lines are oriented away from you.
(c) Both the horizontal and vertical lines are oriented away from you.
(d) Both the horizontal and vertical lines are oriented towards you.

Part- B (2 x 10 = 20 Marks)

11 **a. i.** What is an electrochemical cell? Explain with an example and neat diagram. Give cell notation and reactions involved for a given example. (10 Marks)

OR

b. i. Calculate the standard free-energy change ($\Delta G°$) at 25°C for the reaction
$H_{2(g)} + O_{2(g)} \rightleftharpoons H_2O_{2(l)}$
At 25°C, the standard enthalpy change ($\Delta H°$) is −187.78 kJ/mol, and the absolute entropies of the products and reactants are:

$S°(H_2O_2) = 109.6$ J/(mol.K),
$S°(O_2) = 205.2$ J/(mol.K), and
$S°(H_2) = 130.7$ J/(mol.K).

Is the reaction spontaneous as written? What does ΔG say about the rate of this reaction? (6 Marks)

ii. Compare and contrast wet and dry corrosion. (4 Marks)

12 **a. i.** What is isomerism? How is it classified? Write a note on geometrical isomerism. (6 Marks)

ii. Draw Sawhorse projections for the eclipsed and staggered conformations of ethane. Which of these conformations is more stable and why? (4 Marks)

OR

b. i. Derive the expression and explain the significance of Helmholtz function. (6 Marks)

ii. What are intensive and extensive properties? Give two examples of each. (4 Marks)

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Set- 2

INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.	Date	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	Time	: 8.00-9.00am
Year & Sem.	: I Year & I Sem.	Max. Marks	: 30 marks

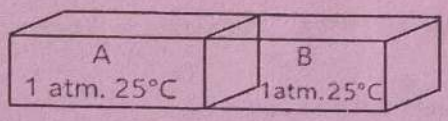
Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part-A (10X1 = 10Marks)
Answer ALL the questions

1. The equilibrium constant for a cell reaction, $Cu_{(g)} + 2Ag^{+}_{(aq)} \leftrightarrow Cu^{2+}_{(aq)} + 2Ag_{(s)}$ is 4×10^{16}. Find $E^{\circ}_{(cell)}$ for the cell reaction.
   (a) 0.63 V (b) 0.49 V (c) 1.23 V (d) 3.24 V

2. An isolated box, equally partitioned, contains two ideal gasses A and B as shown. When the partition is removed, the gases mix. The changes in enthalpy (ΔH) and entropy (ΔS) in the process respectively are:



   (a) Zero, positive (b) Zero, negative (c) Positive, zero (d) Negative, zero

3. Passivity is not reason for inertness of the following:
   (a) Au (b) Al (c) Ti (d) Ni

4. In a reaction, the change in entropy is given as 2.4 cal/K and the change in Gibbs free energy is given as 3.4 kcal, then the change in heat at the temperature of 20 °C is:
   (a) 4.1 cal (b) 3.4 cal (c) 3.4 Kcal (d) 4.1 Kcal

5. Mercury covered by a layer of mercurous chloride in contact with saturated potassium chloride solution is a description of which of the following types of electrodes?
   (a) Chlorine (b) Potassium (c) Calomel (d) Silver/Silver chloride

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6. A process is carried out at constant volume and at constant entropy. It will be spontaneous if:
(a) $\Delta H < 0$ (b) $\Delta U < 0$ (c) $\Delta A < 0$ (d) $\Delta G < 0$

7. The entropy of an isolated system always ____ and reaches ____ when equilibrium is reached.
(a) remains constant, maximum (b) decreases, minimum
(c) increases, maximum (d) decreases, constant

8. The isomers which can be inter converted through rotation around a single bond are termed as:
(a) Tautomers (b) positional isomers (c) conformers (d) diastereomers

9. Which of the following statements is **not** correct?
(a) Fischer projection represents the molecule in an eclipsed conformation
(b) Newman projection can be represented in eclipsed, staggered and skew conformations
(c) Fischer projection of the molecule is its most stable conformation
(d) In Sawhorse projections, the lines are inclined at an angle of 120° to each other

10. The reactivity order of alkyl halides in SN^1 mechanism is
(a) $CH_3X > 1^o > 2^o > 3^o$ (b) $CH_3X < 1^o < 2^o < 3^o$
(c) $CH_3X > 3^o > 2^o > 1^o$ (d) $CH_3X > 3^o > 1^o > 2^o$

Part-B (2 x 10 = 20 Marks)

11(a) (i) Describe the construction and working of an electrochemical cell, provide an example to illustrate its operation. (6 Marks)

(ii) Calculate the entropy change for methane combustion from the given thermodynamic data. The Standard entropies of the substances involved in the above reaction are: ΔS°(J/K. mol)) of $CH_{4(g)}$ =186, $O_{2\ (g)}$ = 205 $CO_{2\ (g)}$ = 214 and $H_2O_{\ (l)}$ = 70 (4 Marks)

(OR)

11(b)
(i) Derive Nernst equation and explain its significance in electrochemistry. (6 Marks)

(ii) Define entropy and explain its significance in thermodynamic and natural processes. (4Marks)

12(a) Explain mechanisms of dry corrosion and wet corrosion, providing the chemical reaction involved in each process. (10 Marks)

(OR)

12(b)
(i) Explain the addition of HBr to propylene, in the presence and absence of peroxide. (8 Marks)

(ii) Define diastereomers with an example. (2 Marks)

No.	BL	CO	PO
1	2	2	2
2	4		2
3	1		1
4	2		2
5	1		3
6	4		2
7	2		1
8	3	3	2
9	2		3
10	3		4

Q. No.	BL	CO	PO
11a. (i)	1	2	1
(ii)	2		2
11b. (i)	3		3
(ii)	1		1
12a.	2		2
12b. (i)	3	3	3
(ii)	2		4

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Set-2

ANSWER KEY-Internal Assessment - II [CLA1]

Program: B.Tech	Date: 17/10/2023
Course Code & Title:21CYB101J & Chemistry	Duration: 8.00 – 9.00 am
Year &Sem: 1 Year & 1 Sem	Max. Marks: 30 marks

Part – A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. **(d) Ionization energy of oxygen is higher than that of nitrogen**
2. **(c) F < Cl > Br > I**
3. **(b) Decreases**
4. **(a) Zn(II), K(I), Fe (III)**
5. **(a) Zero, positive**
6. **(d) 4.1 Kcal**
7. **(b) Size of the electrode**
8. **(c) Calomel**
9. **(d) All of the above**
10. **(b) Co**

Part – B (2 x 10 = 20 Marks)

11. a. i Define Ionization energy, discuss factors affecting it and its variation across the period, down the group in the periodic table. (10 marks)

Ans:
Definition – 2 marks,
Factors affecting (any 3)- 6 marks,
Variation across period – 1 mark,
down the group – 1 mark

(OR)

b. i. Calculate the change in the enthalpy of the following reaction if the heat of formations of $H_2O_{2(l)}$ and $H_2O_{(l)}$ are -188 kJ/mol and -286 kJ/mol respectively.

$2H_2O_{2(l)} \rightarrow 2H_2O_{(l)} + O_{2\,(g)}$ (4 Marks)

Ans: Equation: 1 mark, Calculation -3 marks

$\Delta H_{rec} = \Delta H_{prod.} - \Delta H_{react.}$
$= [2(-286) + 0] - 2(-188)$
$= -196$ kJ/mol

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ii. Derive expression for Helmholtz Free energy and give its significance. (6 marks)

Ans:

Derivation- 4 marks, Significance – 2 marks

12 a. i For a Daniel Cell involving a cell reaction, $Zn_{(s)} + Cu^{2+}_{(aq)} \leftrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$
The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}_{(aq)}$, $Zn^{2+}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (6 Marks)

Ans:

$\Delta G^{o}_{rec} = \Delta G^{o}_{f(prod)} - \Delta G^{o}_{f(react)}$

$= (-154 + 0) - (0 + 64.4) = -218.4$ kJ/mol

$E^{o} = -\Delta G^{o}_{rec}/nF$

$= 1.13$ V

Calculation of ΔG^{o}_{rec} – 3marks
Calculation of EMF – 3 marks

ii Define entropy and give its physical significance. (4 Marks)
Ans: Definition and expression – 1.5 mark
Significance – 1.5 marks

(OR)

b. i. Write a note on Pourbaix diagram of Fe. (10 Marks)

Ans:
Labelled diagram with 5 curves and regions – 3 marks, equations – 3 marks, Significance – 2 marks, Features and limitations – 2 marks

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Set- 1

INTERNAL ASSESSMENT – II [FJ-II]

Program	: B. Tech.	Date	: 21-11-2024
Course Code & Title	: 21CYB101J & Chemistry	Time	: 8.00-9.00am
Year & Sem.	: 1 Year & 1 Sem.	Max. Marks	: 30 marks

Course Articulation Matrix

At the end of this course, learners will be able to:	POs				
Course Outcomes (CO)	1	2	3	4	5
CO1 Rationalize bulk properties using periodic properties of elements, evaluate water quality parameters like hardness and alkalinity	3		3	2	
CO2 Utilize the concepts of thermodynamics in understanding thermodynamically driven chemical reactions, determine acidic strength and redox potentials of aqueous solution	3	3	3		
CO3 Perceive the importance of stereochemistry in synthesizing organic molecules applied in pharmaceutical industries, determine acidic strength and conductance of aqueous solution		3	3	2	
CO4 Utilize the concepts of polymer processing for various technological applications, determine average molecular weight of the polymer	3		3	3	
CO5 Analyze the importance of advanced processing techniques towards engineering applications and measure the acidic strength of aqueous solution	3		3		3

Part-A (10X1 = 10 Marks)
Answer ALL the questions

1. When equilibrium is reached inside the two half-cells of the electrochemical cell, what is the net voltage across the electrodes?
   (a) > 1 (b) < 1 (c) = 0 (d) Not defined
2. What will be the Gibbs free energy for the reaction of conversion of ATP into ADP at 293 Kelvin, if the change in enthalpy is 19.07 Kcal and the change in entropy is 90 cal per Kelvin.
   (a) 7.3 cal (b) -5.3 cal (c) 7.3 Kcal (d) -7.3 Kcal
3. An oxide layer is considered as protective if ratio of volume of oxide layer to volume of base metal is
   (a) > 1 (b) > 10 (c) < 1 (d) > 3
4. Which of the following factors does not affect the electrode potential of an electrode?
   (a) Nature of the electrode (metal) (b) Temperature of the solution
   (c) Molarity of the solution (d) Size of the electrode
5. Which of the following statement is incorrect?
   (a) At constant pressure, $\Delta H = \Delta E + P\Delta V$
   (b) Thermodynamic symbol for entropy is S.
   (c) Gibbs free energy is a state function.
   (d) For an endothermic process, ΔH is negative.
6. Rusting of iron in acidic aqueous solution of electrolyte occurs with the evolution of _____ gas.
   (a) Nitrogen (b) Chlorine (c) Oxygen (d) Hydrogen

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7. According to the convention, the Daniel cell is represented as:
(a) Zn | $ZnSO_4$ || $CuSO_4$ | Cu, E = 1.09 volt
(b) Zn | $ZnSO_4$ || Cu | $CuSO_4$, E = 1.09 volt
(c) $ZnSO_4$ | Zn || $CuSO_4$ | Cu, E = 1.09 volt
(d) Zn | ZnS || $CuSO_4$ | Cu, E = 1.09 volt

8. What type of reaction takes place upon treatment of a ketone with HCN to form a cyanohydrin?
(a) Nucleophilic addition (b) Nucleophilic substitution
(c) Electrophilic addition (d) Electrophilic substitution

9. How many aromatic isomers of dibromobenzene exist?
(a) 2 (b) 3 (c) 4 (d) 6

10. Which of the following is true about Fischer Projection?

(a) The vertical lines are oriented away from you and the horizontal lines are oriented toward you.
(b) The vertical lines are oriented towards you and the horizontal lines are oriented away from you.
(c) Both the horizontal and vertical lines are oriented away from you.
(d) Both the horizontal and vertical lines are oriented towards you.

Part-B (2 x 10 = 20 Marks)

11(a) (i) Derive the expression for the Helmholtz free energy and explain its physical significance. (6 Marks)

(ii) For a Daniel Cell involving a cell reaction,

$$Zn_{(s)} + Cu^{2+}_{(aq)} \leftrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$$

The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}_{(aq)}$, $Zn^{2+}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (4 Marks)

(OR)

11(b) (i) Describe potentiometric redox titration with an example and explain its advantages. (6 Marks)

(ii) Differentiate between dry corrosion and wet corrosion. (4 Marks)

12(a) Write a note on the Pourbaix diagram of iron (Fe). (10 Marks)

(OR)

12(b) (i) Explain the SN^1 mechanism with a suitable example. (6 Marks)

(ii) Draw the Newman and Sawhorse projections for the eclipsed and staggered conformations of ethane. Which conformation is more stable and why? (4 Marks)

No.	BL	CO	PO
1	2	2	2
2	4		2
3	1		1
4	2		2
5	1		3
6	4		2
7	2		1
8	3	3	2
9	2		3
10	3		4

Q. No.	BL	CO	PO
11a. (i)	1	2	1
(ii)	2		2
11b. (i)	3		3
(ii)	1		1
12a.	2		2
12b. (i)	3	3	2
(ii)	2		3

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Set-1

ANSWER KEY-Internal Assessment - II [CLA1]

Program: B.Tech	**Date: 17/10/2023**
Course Code & Title: 21CYB101J & Chemistry	**Duration: 8.00 – 9.00 am**
Year & Sem: 1 Year & 1 Sem	**Max. Marks: 30 marks**

Part – A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. **(c) Li < B < Be < C < O < N < F < Ne**
2. **(b) Mn, Zn**
3. **(b) High ionisation potential**
4. **(c) $[AlF_6]^{3-}$**
5. **(a)Exothermic**
6. **(d) -7.3 Kcal**
7. **(c) = 0**
8. **(b) 0.49 V**
9. **(a)> 1**
10. **(a) Au**

Part – B (2 x 10 = 20 Marks)

11. a. i Explain why CsF reacts LiI to give LiF and CsI? (6 Marks)

Ans:

Combination of Soft-Soft and Hard-Hard species is a stable molecule. Li^+ is a hard acid, I^- is a soft base, Cs^+ is soft acid and F^- is hard base. So, the combination of Li^+ and F^- gives stable molecule LiF. Similarly, the combination of Cs^+ and I^- gives a stable molecule CsI.

LiI (Hard-Soft) + CsI (Soft-Hard) → LiF (Hard-Hard) + CsI (Soft-Soft).

Identification of HA, HB, SA, SB – 2 marks
HSAB Principle – 2 marks
Reaction with explanation – 2 marks

ii Arrange M, M^+ and M^- in increasing order of atomic/ionic size and justify the order. (4 Marks)

Ans: $M^+ < M < M^-$
Correct order - 1 mark
Explanation based on effective nuclear charge – 3 marks

(OR)

b. i. Calculate the total entropy change for the transition at 368 K of 1 mol of sulphur from the monoclinic to the rhombic solid state. Given $\Delta H = -401.7$ J mol $^{-1}$ for the transition.

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Assume the surrounding to be an ice-water bath at 0 °C. (4 Marks)

Ans:

The entropy change for phase transition (for 1 mol here) is given as:
$\Delta S(system) = \Delta H_{trans} / T = 1 \times -401.7/368 = -1.09\ JK^{-1}$
The heat given out by system is absorbed by ice-water bath at 273 K
$\therefore \Delta S(surrounding) = 1 \times 401.7/273 = 1.47 JK^{-1}$

so, $\Delta S(total) = \Delta S_{sys} + \Delta S_{surr} = -1.09 + 1.47 = 0.38 JK^{-1}$

ii. Explain construction and working of an electrochemical cell with an example. (6 Marks)

Ans:
Labelled diagram: 2.5 marks
Working with redox reactions – 2.5 marks
Cell notation or EMF equation: 1 mark

12 a. i Four identical containers have equal amounts of helium gas that all start at the same initial temperature. Containers of gas also have a tightly fitting movable piston that does not allow any of the gas to escape. Each sample of gas is taken through different process as described below:
Sample 1: 500 J heat exits the gas and the gas does 300 J of work
Sample 2: 500 J of heat enters the gas and the gas does 300 J of work
Sample 3: 500 J of heat exits the gas and 300 J of work is done on the gas
Sample 4: 500 J of heat enters the gas and 300 J of work is done on the gas
Arrange the samples in increasing order of their temperature. (6 Marks)

Ans: Whichever gas has the largest increase in internal energy ΔU will also have the greatest increase in temperature ΔT (since temperature and internal energy are proportional). Using first law of thermodynamics for each process,
Process 1: $\Delta U = -800$ J
Process 2: $\Delta U = +200$ J
Process 3: $\Delta U = -200$ J
Process 4: $\Delta U = +800$ J
$\Delta U_4 > \Delta U_2 > \Delta U_3 > \Delta U_1$ and $\mathbf{T_4 > T_2 > T_3 > T_1}$
Calculation of ΔU for 4 processes – 4 marks
Order of internal energy and temperature – 2 marks

ii Differentiate between dry corrosion and wet corrosion. (4 Marks)

Ans: Any 4 points – 4 marks

(OR)

b. i. Describe potentiometric redox and potentiometric acid base titration (10 Marks)

Ans: Redox titration- 5 marks, Acid base titration – 5 marks
Explanation with example including working and reference electrode, reactions involved, cell notation, expression for emf and 3 plots)

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Set-2

INTERNAL ASSESSMENT – II [CLA1 T2]

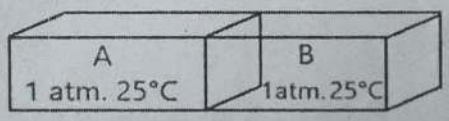
Program: B.Tech	**Date: 17/10/2023**
Course Code & Title:21CYB101J & Chemistry	**Time: 8.00-9.00AM**
Year &Sem: I Year & I Sem	**Max. Marks: 30 marks**

Part – A (10 x 1 = 10 Marks)

Answer ALL the Questions

1. The incorrect statement about the ionization energy is -
   (a) Ionization energy increases for each successive electron
   (b) Noble gases have the highest ionization energy
   (c) A big jump in ionization energy indicates a stable configuration
   (d) Ionization energy of oxygen is higher than that of nitrogen .
2. Among halogens, the correct order of amount of energy released in electron gain is -
   (a) F > Cl > Br > I
   (b) F < Cl < Br < I
   (c) F < Cl > Br > I
   (d) F < Cl < Br < I
3. On-going from right to left in a period in the periodic table, the electronegativity of the elements
   (a) Increases (b) Decreases (c) Remains unchanged (d) Decreases first then increases
4. Suggest in each of following pairs, which of the ions is harder.
   (i) Zn (II) or Hg (II) (ii) K(I) or Cu (I) (iii) Fe(II) or Fe(III)

   (a) Zn(II), K(I), Fe (III) (b) Hg (II), Cu (I), Fe(II)
   (c) Zn (II), Cu (I), Fe (III) (d) Hg (II), K(I), Fe (III)
5. An isolated box, equally partitioned, contains two ideal gasses A and B as shown. When the partition is removed, the gases mix. The changes in enthalpy (ΔH) and entropy (ΔS) in the process respectively are:



   (a) Zero, positive (b) Zero, negative
   (c) Positive, zero (d) Negative, zero
6. In a reaction, the change in entropy is given as 2.4 cal/K and the change in Gibbs free energy is given as 3.4 kcal, then the change in heat at the temperature of 20 °C is:
   (a) 4.1 cal (b) 3.4 cal (c) 3.4 Kcal (d) 4.1 Kcal
7. Which of the following factors does not affect the electrode potential of an electrode?
   (a) Nature of the electrode (metal) (b) Temperature of the solution
   (c) Molarity of the solution (d) Size of the electrode
8. Mercury covered by a layer of mercurous chloride in contact with saturated potassium chloride solution is a description of which of the following types of electrodes?

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(a) Chlorine (b) Potassium (c) Calomel (d) Silver/Silver chloride

9. If an alloy of Au and Zn is immersed in HCl, which among the following will be feasible?
   (a) H_2 evolution (b) Zn dissolution
   (c) Formation of nonporous Au (d) All of the above
10. When Pt and Co are electrically connected, which one gets corroded?
   (a) Pt (b) Co (c) None (d) Can't decide

Part – B (2 x 10 = 20 Marks)

11. a. Define Ionization energy, discuss factors affecting it and its variation across the period, down the group in the periodic table. (10 marks)

(OR)

b. i. Calculate the change in the enthalpy of the following reaction if the heat of formations of $H_2O_{2(l)}$ and $H_2O_{(l)}$ are -188 kJ/mol and -286 kJ/mol respectively.

$2H_2O_{2(l)} \rightarrow 2H_2O_{(l)} + O_{2\,(g)}$ (4 Marks)

ii. Derive expression for Helmholtz Free energy and give its significance. (6 marks)

12. a. i. For a Daniel Cell involving a cell reaction, $Zn_{(s)} + Cu^{2+}{}_{(aq)} \leftrightarrow Zn^{2+}{}_{(aq)} + Cu_{(s)}$
The standard free energy of formation of $Zn_{(s)}$, $Cu_{(s)}$, $Cu^{2+}{}_{(aq)}$, $Zn^{2+}{}_{(aq)}$ are 0, 0, 64.4kJ/mol and -154 kJ/mol respectively. Calculate the standard EMF of the cell. (6 Marks)

ii. Define entropy and give its physical significance. (4 Marks)

(OR)

b. With a neat sketch of Pourbaix diagram for Iron system, explain redox and acid-base reactions. (10 Marks)

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SRM

DEPARTMENT OF CHEMISTRY **SET 1**
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

INTERNAL ASSESSMENT – II

Program: B. Tech.	Date: 12-11-2022
Course Code & Title: 21CYB101J & Chemistry	Duration: 12.30-01.30 PM
Year & Semester: I Year & I Semester	Max. Marks: 30 Marks

Part- A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. The change of enthalpy in an endothermic reaction is
   (a) positive (b) negative (c) constant (d) can be any of the above

2. Which of the following is an **incorrect** statement?
   (a) Corrosion is an irreversible process
   (b) Corrosion is a non-spontaneous process
   (c) Corrosion is a degradation process
   (d) Corrosion is a spontaneous process

3. Rusting of iron in acidic aqueous solution of electrolyte occurs with the evolution of _____gas.
   (a)Nitrogen
   (b)Chlorine
   (c)Oxygen
   (d) Hydrogen

4. A system absorbs 50 J of heat and does 11 J of work in a process. The system follows a different thermodynamic path between the same initial and final states and does 15 J work, the heat transferred in the process is
   (a) 39 J (b) 65 J (c) 54 J (d) 46 J

5. Which of the following is **false** regarding galvanic cells?
   (a) It converts chemical energy into electrical energy
   (b) The electrolytes taken in the two beakers are different
   (c) The reactions taking place are non-spontaneous
   (d) To set up this cell, a salt bridge is used

6. ΔG for the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ is
   (a) –FE (b) FE (c) 2FE (d) -2FE

7. For which of the following conditions a reaction will be spontaneous at all the temperature?
   (a) $\Delta H < 0$ and $\Delta S < 0$
   (b) $\Delta H < 0$ and $\Delta S = 0$
   (c) $\Delta H < 0$ and $\Delta S > 0$

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(d) both (b) and (c)

8. Which of the following is the application of Pourbaix diagrams?
(a) Predicting the spontaneous direction of reactions
(b) To reduce corrosion attack
(c) To estimate the composition of corrosion product
(d) All of the above

9. Which of the following statements is **not** correct?
(a) Fischer projection represents the molecule in an eclipsed conformation
(b) Newman projection can be represented in eclipsed, staggered and skew conformations
(c) Fischer projection of the molecule is its most stable conformation
(d) In Sawhorse projections, the lines are inclined at an angle of 120° to each other

10. The compounds $C_2H_5OC_2H_5$ and $CH_3OCH_2CH_2CH_3$ are
(a) chain isomers
(b) geometrical isomers
(c) metamers
(d) conformational isomers

Part- B (2 x 10 = 20 Marks)

11 **a. i.** What is Nernst Equation? Derive its expression and describe one application of Nernst equation in detail. (10 Marks)

OR

b. i. The cell in which the following reaction occurs:

$$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + I_{2\,(aq)}$$

has $E^\circ_{cell} = 0.236V$ at 298 K. Calculate the standard Gibbs free energy and the equilibrium constant of the reaction. (Given: F = 96,485 C/mol) (6 Marks)

ii. Differentiate between Dry Corrosion and Wet corrosion (4 Marks)

12 **a. i.** What is Gibb's free energy? Derive its expression. (6 Marks)

ii. Define entropy and give its significance. (4 Marks)

OR

b. i. Write a note on structural isomerism in organic compounds. (6 Marks)

ii. Draw Newman projections for the eclipsed and staggered conformations of ethane. Which of these conformations is more stable and why? (4 Marks)

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SET 1

SRM

DEPARTMENT OF CHEMISTRY
College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

INTERNAL ASSESSMENT – II-Answer Key

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Semester: I Year & I Semester

Date: 19/11/2022
Duration: 12.30 – 1.30 pm
Max. Marks: 30 Marks

Part- A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. The change of enthalpy in an endothermic reaction is
   (a) positive
2. Which of the following is an **incorrect** statement?
   (b) Corrosion is a non-spontaneous process
3. Rusting of iron in acidic aqueous solution of electrolyte occurs with the evolution of _____gas.
   (d) Hydrogen
4. A system absorbs 50 J of heat and does 11 J of work in a process. The system follows a different thermodynamic path between the same initial and final states and does 15 J work, the heat transferred in the process is
   (c) 54 J
5. Which of the following is **false** regarding galvanic cells?
   (c) The reactions taking place are non-spontaneous
6. ΔG for the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ is
   (d) -2FE
7. For which of the following conditions a reaction will be spontaneous at all the temperature?
   (d) both (b) and (c)
8. Which of the following is the application of Pourbaix diagrams?
   (d) All of the above
9. Which of the following statements is **not** correct?
   (c) Fischer projection of the molecule is its most stable conformation
10. The compounds $C_2H_5OC_2H_5$ and $CH_3OCH_2CH_2CH_3$ are
    (c) metamers

Part- B (2 x 10 = 20 Marks)

11 a. i. What is Nernst Equation? Derive its expression and describe one application of Nernst equation in detail. (10 Marks)

The Nernst equation defines a relation between the cell potential of an electrochemical cell, the standard cell potential, temperature, and the reaction quotient. It is used for cell potential determination under non-standard conditions. **(1 Mark)**

Derivation of expression **(5 marks)**

Description of any one application **(4 Marks)**

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OR

b. i. The cell in which the following reaction occurs:

$$2Fe^{3+}_{(aq)} + 2I_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + I_{2(aq)}$$

has $E^{\circ}_{cell} = 0.236V$ at 298 K. Calculate the standard Gibbs free energy and the equilibrium constant of the reaction. (Given: F = 96,485 C/mol) (6 Marks)

Solution: The half cell reactions can be written as:

$$Anode: \ 2I^{-}_{(aq)} \rightarrow I_{2(s)} + 2e$$

$$Cathode: \ 2Fe^{3+}_{(aq)} + 2e \rightarrow 2Fe^{2+}_{(aq)}$$

$$Number\ of\ electrons\ exchanged,\ n = 2$$

Here, $n = 2$, $E^{\theta}_{cell} = 0.236$ T = 298 K

We know that:

$$\triangle_r G^{\theta} = -nFE^{\theta}_{cell}$$

$$= -2 \times 96487 \times 0.236$$

$$= -45541.864\ J\ mol^{-1}$$

$$= -45.54\ kJ\ mol^{-1}$$

Writing half- cell reactions and calculating 'n' (2 Marks)
Calculation of Free energy (4 marks)

ii. Differentiate between Dry Corrosion and Wet corrosion (4 Marks)
Any 4 points of difference (each 1 mark)

12 a. i. What is Gibb's free energy? Derive its expression. (6 Marks)
Definition and explanation (2 Marks)
Derivation (4 Marks)

ii. Define entropy and give its significance. (4 Marks)
Definition (1 Mark)- Any two significance (3 Marks)

OR

b. i. Write a note on structural isomerism in organic compounds. (6 Marks)
Classification into (1) Chain isomerism (2) Position Isomerism (3) Functional group isomerism and (4) Metamerism with one example (each 1.5 marks)

ii. Draw Newman projections for the eclipsed and staggered conformations of ethane. Which of these conformations is more stable and why? (4 Marks)
Drawing of eclipsed and staggered conformation (3 Marks)
More stable conformation and its reason (1 Mark)

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INTERNAL ASSESSMENT – II

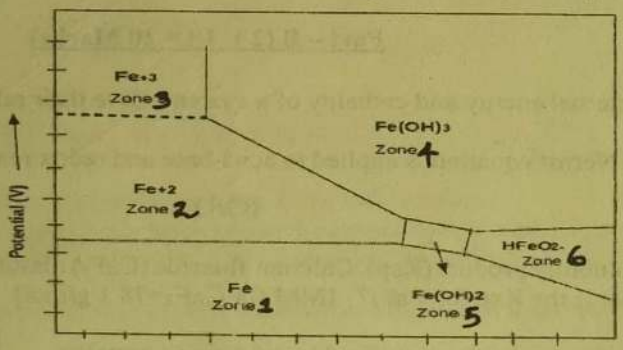
Set-1

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

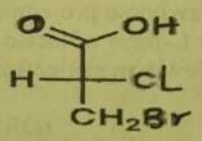
Date: 27-03-2023
Time: 12.30 – 1.30 PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10Marks)
Answer ALL The Questions

1. All the naturally occurring processes proceed spontaneously in a direction that leads to
   a. decrease of entropy b. increase of enthalpy c. an increase of free energy d. decrease of free energy
2. If spontaneity of a reaction is temperature dependent and endothermic, then
   a. $\Delta H<0$ and $\Delta S>0$ b. $\Delta H<0$ and $\Delta S<0$ c. $\Delta H>0$ and $\Delta S>0$ d. $\Delta H>0$ and $\Delta S<0$.
3. The e.m.f and the standard e.m.f of a cell in the following reaction is 5 V and 5.06 V at room temperature, $Ni(s) + 2Ag^{+}_{(a)} \rightarrow Ni^{2+}(0.02M) + 2Ag_{(s)}$. What is the concentration of Ag^{+} ions?
   a) 0.0125 M b) 0.0314 M c) 0.0625 M d) 0.0174 M
4. Metals do not exist in nature in the form of-----
   a) Nitrates b) Sulphates c) Carbonates d) Oxides
5. Which of the following zone of the given Pourbaix diagram indicates the passive zone?



   a) Zone 1 and 2 b) Zone 2 and 3 c) Zone 2, 3 and 6 d) Zone 4 and 5
6. Assign R/S notation for the given compound:



   a) R b) S c) Neither R nor S d) R or S

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7. Diastereomers are

a) Geometrical isomers b) Mirror images c) Non-mirror images d) Unstable molecules

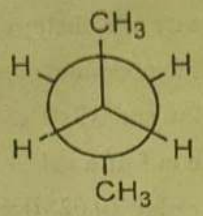
8. The water molecule exhibits .....rotation axis of symmetry.

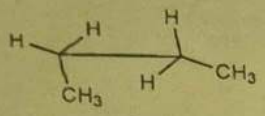
a) C_3 b) C_2 c) C_4 d) C_n

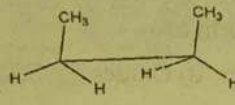
9. The potential energy of n-butane is maximum for _____

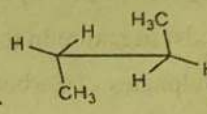
a) Skew conformations b) Staggered conformations c) Eclipsed conformations d) Gauche

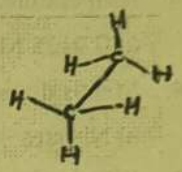
10. Which of the following sawhorse representations is correct for the given Newman projection?



a. 

b. 

.c. 

d. 

Part – B (2 x 10 = 20 Marks)

11. a. i. Define internal energy and enthalpy of a system. Give their relation. (4 Marks)

ii. How the Nernst equation is applied to acid-base and redox reactions. (6 Marks)

(OR)

b. i. Define solubility product (Ksp). Calcium fluoride (CaF_2) dissolves in water to the extent of 0.00170 g/100ml. What is the Ksp for CaF_2? [MM for CaF_2=78.1 g/mol] (4 Marks)

ii. Explain the mechanism involved in oxidation corrosion. (6 Marks)

12. a. i. Give the differences between mesomers and racemic mixture. (4Marks)

ii. Brief on the projections and sketch it for the following compounds. (6 Marks)

A. Staggered form of Sawhorse projection for 2-butanol.
B. Fisher projection for L-form of lactic acid.
C. Staggered form of Newman projection for CH_3CH_3.

(OR)

b. Explicate structural isomerism exhibited by organic com pounds with suitable examples.

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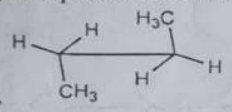
Set-1

INTERNAL ASSESSMENT – II
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 27.03.2023
Time: 12.30 – 1.30 PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10Marks)

1. d. decrease of free energy
2. c. $\Delta H>0$ and $\Delta S>0$
3. d) 0.0174 M
4. a) Nitrates
5. d) Zone 4 and 5
6. b) S
7. c) Non-mirror images
8. b) C_2
9. c) Eclipsed conformations
10. c. 

Part – B (2 x 10 = 20 Marks)

11. a. i. Define internal energy and enthalpy of a system. Give their relation. (4 Marks)

The sum of all the energies contained in a system, such as mechanical, chemical, potential, kinetic, and so on, is its internal energy. Internal energy is denoted by the letter U, and joules are the unit of measurement (J).
Enthalpy is a measure of the total heat energy, absorbed or released, in a thermodynamic system. [2]
Relation between U and H. [2]

ii. How the Nernst equation is applied to acid-base and redox reactions. (6 Marks)

Application towards acid-base and redox reactions using Nernst equation. (3x2=6 Marks)

(OR)

b. i. Define solubility product (Ksp). Calcium fluoride (CaF_2) dissolves in water to the extent of 0.00170 g/100ml. What is the Ksp for CaF_2? [MM for CaF_2=78.1 g/mol] (4 Marks)

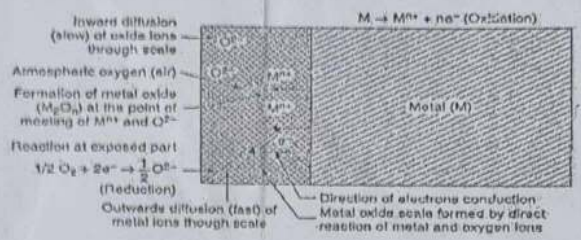
Definition. (1 Mark)

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Solubility of $CaF_2 = 1.7 \times 10^{-3} g/100mL = 17 \times 10^{-3} gL^{-1}$
Molar mass of $CaF_2 = (40 + 2 \times 19) = 78 gmol^{-1}$
Solubility of $CaF_2 = \frac{(17 \times 10^{-3} gL^{-1})}{(78.0 gmol^{-1})} = 2.18 \times 10^{-4} molL^{-1}$
The solubility equilibrium for the saturated solution of CaF_2 is :
$CaF_2(s) \overset{aq}{\Leftrightarrow} Ca^{2+}(aq) + 2F^-(aq)$
$[Ca^{2+}(aq)] = 2.18 \times 10^{-4} molL^{-1}$
$[F^-(aq)] = 2 \times 2.18 \times 10^{-4} molL^{-1}$
Applying Law of chemical equilibrium :
$K_{sp} = [Ca^{2+}(aq)][F^-(aq)]^2$
$= (2.18 \times 10^{-4} molL^{-1}) \times (2.18 \times 10^{-4} molL^{-1})^2 = 4.14 \times 10^{-11} mol^3.$

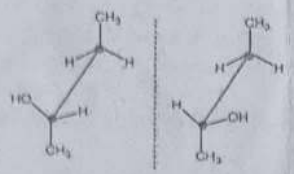
(3 Marks)

ii. Explain the mechanism involved in oxidation corrosion. (6 Marks)
Mechanism with equations and explanation. **(5)**
Diagram. **(1)**

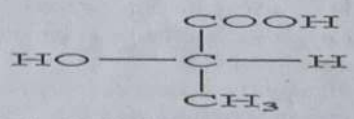


12. a. i. Give the differences between mesomers and racemic mixture. (4Marks)
Any 2 differences with examples. **(2x2=4Mks)**

ii. Brief on the projections and sketch it for the following compounds. (6 Marks)
Brief on each projection. **(3 Marks)**
Projections **(3 Marks)**



A. 2-butanol



B. L(+) lactic acid



C. CH_3-CH_3

(OR)

b. Explicate structural isomerism exhibited by organic compounds with suitable examples.
Explanation on all five isomerism with examples. **(5x2=10 Marks)**

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Department of Chemistry
SRM Institute of Science and Technology (SRM IST)
College of Engineering & Technology

Surprise Test

Subject: Chemistry
Subject code: 21CYB101J
Branch: B.Tech (CSE - C1 Section)

Duration: 12.30-1.30AN
Marks: 20

SET - 2

Answer any four questions **(4 x 5 = 20)**

1. Write a note on free energy for corrosion reaction.
2. What is entropy? Explain its significance.
3. Explain the mechanism of wet corrosion with example.
4. Define the following with examples.
   a) Plane of symmetry
   b) Centre of symmetry
5. Write types of stereoisomerism with an examples.
6. A reaction at equilibrium releases 50 Joules of heat energy at 37 °C. What is the entropy change for this reaction?

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Department of Chemistry
SRM Institute of Science and Technology (SRM IST)
College of Engineering & Technology

Surprise Test

Subject: Chemistry
Subject code: 21CYB101J
Branch: B.Tech (CSE - C1 Section)

Duration: 12.30-1.30AN
Marks: 20

SET - 1

Answer any four questions **(4 x 5 = 20)**

1. What is the relation between enthalpy and Internal energy?
2. Discuss on the mechanism of electrochemical corrosion with an examples.
3. What is an Electrochemical cell or Galvanic cell? Give its representation.
4. Define the terms i. Entropy ii. Enthalpy
5. Write any three types of structural isomerism with an examples.
6. One mole of an ideal gas expands against a constant external pressure of 1 atm from a volume of 20 dm^3 to a volume of 70 dm^3. Calculate work done by the gas in joules.

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DEPARTMENT OF CHEMISTRY
FACULTY OF ENGINEERING AND TECHNOLOGY
SRM IST, KATTANKULATHUR

SURPRISE TEST

Program: B. Tech
Sub. Code: 21CYB101J
Sub. Title: Chemistry
Semester: IV

Max. Marks: 20
Time: 8.50-9.40 am
Date: 19.03.2024

SET-II

Answer any four questions (4 x 5= 20 Marks)

1. Derive Nernst equation?
2. Explain construction and working of an electrochemical cell with an example
3. Write a neat sketch of Pourbaix diagram for Iron system, Explain redox and acid base reactions
4. Explain the various conformations of n-butane in detail
5. The cell in which the following reaction occurs:
   $2Fe^{3+}(aq) + 2I^{-}(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$
   has E_o cell = 0.236 V at 298 K.
   Calculate the standard Gibbs free energy and the equilibrium constant of the reaction.
   (Given: F = 96,485 C/mol)

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DEPARTMENT OF CHEMISTRY
FACULTY OF ENGINEERING AND TECHNOLOGY
SRM IST, KATTANKULATHUR

SURPRISE TEST

Program: B. Tech
Sub. Code: 21CYB101J
Sub. Title: Chemistry

Max. Marks: 20
Time: 8.50-9.40 am
Date: 19.03.2024

SET-I

Answer any four questions (4 x 5= 20 Marks)

1. Derive Gibbs- Helmholtz equation?
2. Define and give its physical significance of entropy and Gibbs free energy.
3. Explain Cahn- Ingold Prelog rules for the determination of absolute configuration
4. Describe the oxygen absorption type of electrochemical corrosion with a suitable mechanism
5. Calculate the standard Gibbs free energy and the equilibrium constant of the reaction

   $Zn\ (s) + Cu^{2+}\ (aq) \longrightarrow Zn^{2+}\ (aq) + Cu\ (s)$.

   Given : E° for $Zn^{2+}/Zn = -0{\cdot}76$ V and

   E° for $Cu^{2+}/Cu = +0{\cdot}34$ V

   $R = 8{\cdot}314\ JK^{-1}\ mol^{-1}$

   $F = 96500\ C\ mol^{-1}$.

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College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

INTERNAL ASSESSMENT – II-CLA1 T2
RETEST

Program: B. Tech.
Course Code & Title: 21CYB101J & Chemistry
Year & Semester: I Year & I Semester

Date: 08/11/2023
Duration: 12.30 – 1.30 pm
Max. Marks: 30 Marks

Part- A (10 x 1 = 10 Marks)

Answer ALL The Questions

1. Which of the following is correct for the net entropy change in an irreversible process?
   (a) It is positive (b) It is negative (c) It is zero (d) all of the above
2. For a spontaneous process, the change in Gibbs Funtion is equal to.....
   (a) heat content of the system (b) entropy change of the system (c) work of e:pansion (d) useful work
3. caluclate the Gibbs free energy for the reaction of conversion of ATP in to ADP at 293 K in the change in enthalphy is 19.07 Kilo cal and the change in entropy is 90 cal per kelvin (a) 7.3 Cal (b) -5.3 K cal (c) 7.3 K cal (d) -7.3 Kcal
4. During an adiabatic expansion of 3 moles of a gas, the change in internal energy was found to be equal to -100 J. The work done during the process is equal to:
   (a) 122 j (b) 100 J (c) -300 J (d) 300 J
5. Which of the following condition is satisfied when the cell reaction in the electrochemical cell is spontaneous?
   (a) $\Delta G^\circ < 0$ (b) $\Delta G^\circ > 0$ (c) Eo Cell =0 (d) Eo Cell< 0
6. ΔG for the reaction $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ is
   (a) -2FE (b) -1 FE (3) 2 FE (4) 1 Fe
7. In an isothermal process, the internal energy of gas molecules
   (a) Increases (b) decreases (c) remains constant and (d) none of above
8. Which of the following option describes the limitations of Pourbaix diagrams?
   (a) Rate of reaction cannot be predicted (b)It neglects the impurities in working conditions (c) stability of metallic species is predicted (d) Rate of reaction is not predicted and it neglects the impurities of working conditions
9. Anhydrous inorganic liquid metal surface in absence of moisture undergoes _____.
   (a) Wet corrosion (b) Dry corrosion (c) galvanic corrosion (d) Pitting Corrosion
10. Nernst equation for an electrode is based on the variation of electrode potential of an electrode with
    (a) temperature (b) concentration of electrolyte (c) both a and b (d) density of electr

Part- B (2 x 10 = 20 Marks)

11 a. i. What is Nernst Equation? Derive its expression and describe one application of Nernst equation in detail. (10 Marks)

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SRM

DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603203

Set-I

INTERNAL ASSESSMENT – III [CLA1-T3]

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: 1 Year & 1 Sem

Date: 28/11/2023
Time: 12:30 – 1:30 PM
Max. Marks: 30 marks

Part – A (10 x 1 = 10 Marks)

Answer all the questions

1 In most of the world, acetaminophen is best known as which of the following?
   a) Paracetamol
   b) Methamphetamine
   c) Asprin
   d) Naproxen

2. Identify the polymer formed by addition polymerization
   a) Nylon
   b) Polyester
   c) Teflon
   d) Bakelite

3 The synthesis of which of the following polymers involves the repeated loss of small molecules?
   a) Polythene
   b) Buna-S
   c) Buna-N
   d) Nylon-6,6

4 _____undergoes permanent deformation on heating.
   a) Polythene
   b) PVC
   c) Teflon
   d) Bakelite

5 Arrange the following in the decreasing order of leaving group in nucleophilic substitution reaction.
   a) $H^- > Cl^- > HO^- > Br^- > CH_3COO^-$
   b) $Cl^- > Br^- > HO^- > H^- > CH_3COO^-$
   c) $Cl^- > Br^- > CH_3COO^- > HO^- > H^-$
   d) $HO^- > CH_3COO^- > H^- > Br^- > Cl^-$

6 Minimum interplanar spacing required for Bragg's diffraction is
   a) $\lambda/4$ b) $\lambda/2$
   c) 4λ d) 2λ

7 Select the incorrect statement from the following option.
   a) Thermosets have 3-dimensional, cross-linked network structure

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b) Thermosets cannot be remoulded, reused or reclaimed
c) Thermosets are hard, strong and brittle
d) Thermosets are soluble in suitable solvents

8 Which of the following is used for making rechargeable batteries?
a) Nylon
b) Polyester
c) Polyaniline
d) Polyacrylonitrile

9 The dehydration of alcohols is an example of _____
a) E2 reaction
b) S_N2 reaction
c) S_N1 reaction
d) E1 reaction

10 The most suitable reagent for the following transformation is



a) $KMnO_4$
b) $NaBH_4$
c) $K_2Cr_2O_7$
d) $LiAlH_4$

Part – B (2 x 10 = 20 Marks)

11. a. Discuss the addition polymerization process in the synthesis of PVC, Teflon along with the properties and applications. (10 Marks)

(OR)

b. i. Explain polymer tacticity and its types? (6 Marks)
ii. Write a short note on Dieckmann condensation. (4 Marks)

12. a. i. Explain Bragg's law with a neat diagram. (6 Marks)
ii. For the intercepts x, y, and, z with values of 3, 1, and 2 respectively, find the Miller indices. (4 Marks)

(OR)

b. i Explain in detail the conformational analysis of n-butane with potential energy diagram. (5 Marks)
ii. Explain briefly the E1 and E2 reactions with an example for each. (5 Marks)

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